

CHAPTER 2 - Organising and visualising variables

SDS188 - Group 3

2026

Datasets are available on EMSLearn

1 Categorical data

1.1 Example 1

Use the data in the *Categorical 1* sheet for the following questions.

Data: TV shows and movies available on Netflix (2019)

X = Country in which TV show/movie is produced = Australia, Brazil, South Africa, UK, USA

1. Create a summary table for the data.
2. According to the data, in which country are most of the shows produced?
3. What percentage of TV shows are produced in Brazil?
4. Use two suitable graphs to visualise the data.

1.2 Example 2

Use the data in the *Flight Price* sheet for the following questions.

Applicable data:

- *airline*: The name of the airline company
- *stops*: The number of stops between the source and destination cities

1. Create a contingency table for the two relevant variables.
2. Which airline does not stop more than once?
3. Which airline has a greater chance to stop two or more times?
4. What is the probability that a flight that does not stop is from Indigo airline?
5. Draw a side-by-side bar chart of the table.

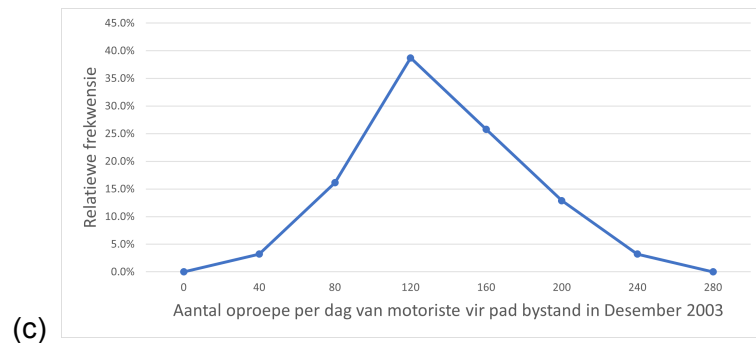
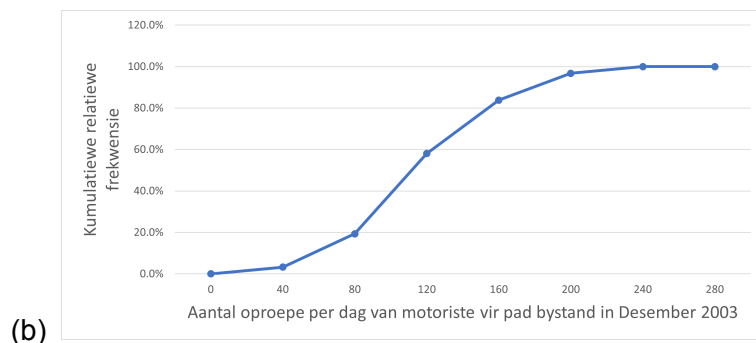
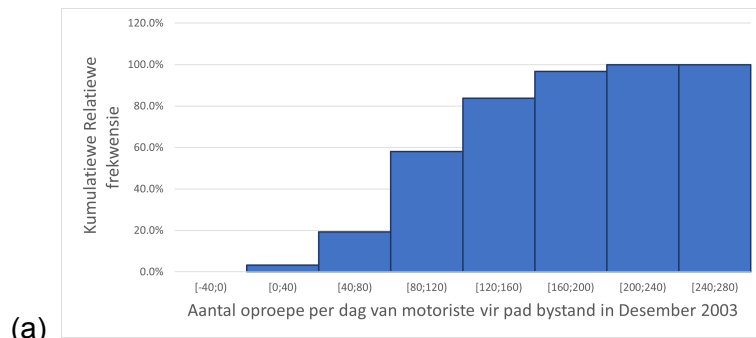
2 Numerical data

2.1 Example 3

Use the data in the *Numerical 1* sheet for the following questions.

Data: Number of calls per day from motorists for roadside assistance in December 2003

1. Sort the data in ascending order.
2. Use a class interval width of 40 and construct a frequency distribution.
3. Where is the data concentrated?
4. For how many days were there less than 160 calls? (Use the cumulative frequency column to obtain a simple answer)
5. Use the percentage column to draw a histogram and a percentage polygon.
6. Which of the following graphs is the correct version of the ogive of the data:



2.2 Example 4

Use the data in the *Numerical 2* sheet for the following questions.

Data: Years experience (X) and salary (Y)

1. Draw a scatter plot of years experience on the X axis and salary on the Y axis.
2. Which of the following statements is the most correct:
 - (a) There is no relationship between years experience and salary. Any change in years experience will have no effect on salary.

- (b) There is a positive relationship between years experience and salary. As years experience increases, salary will also increase.
- (c) There is a negative relationship between years experience and salary. As years experience increases, salary will decrease.

2.3 Example 5

Use the data in the *Power Consumption* sheet for the following data.

Data: Power consumption of some states in India

1. Draw three time series plots of the power consumption of the three states on one graph.
2. Which state's pattern/trend is concealed relative to the other states? Draw two separate graphs, where the first is this state's power consumption over time and the second the other two states' power consumption over time.

3 Mixed data

3.1 Example 6

Use the data in the *Flight Price* sheet for the following questions.

Applicable data:

- *airline*: The name of the airline company
 - *stops*: The number of stops between the source and destination cities
 - *price*: The price of the ticket
1. Adapt the contingency table from 1.2 to show the average ticket prices instead of the frequencies.

2. Obtain the separate datasets for each category of the *stops* variable. (Slide 49). For each dataset, extract the *price* variable and construct a frequency distribution of the ticket prices for flights that don't make stops, flights that make one stop and flights that make two or more stops.

Use the following classes:

Klas	stops: one	stops: two or more	stops: zero
1	(0;20000]	(0;16500]	(0;9000]
2	(20000;40000]	(16500;33000]	(9000;18000]
3	(40000;60000]	(33000;49500]	(18000;27000]
4	(60000;80000]	(49500;66000]	(27000;36000]
5	(80000;100000]	(66000;82500]	(36000;45000]
6	(100000;120000]	(82500;99000]	(45000;54000]
7	(120000;140000]	(99000;115500]	(54000;63000]

3. Draw three percentage polygons on one graph.